# VIHARI REDDY

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## SUMMARY

Master's in Information Technology (3.86/4.0) graduate with hands-on experience building AWS lakehouse and a BigQuery E-commerce platform processing **50M+ events** using Python, SQL and PySpark. Brings professional experience in data quality, business analytics, and data migration. Open to Data Engineer, Data Analyst and Business Intelligence roles

## EDUCATION

**Belhaven University** 01/2025 – 05/2026
*M.S. Information Technology* *Jackson, MS*

**Jawaharlal Nehru Technological University** 08/2016 – 09/2020
*B.S. Computer Science* *Hyderabad, India*

## SKILLS

- **Languages:** SQL, Python, R, Scala
- **Data Engineering:** Dimensional Modeling, ETL, Apache Spark, PySpark, dbt, Pandas, Kafka, Airflow
- **Data Analytics:** Exploratory Analysis, Statistical Analysis, Hypothesis Testing, A/B Testing
- **Data Visualization:** Tableau, Power BI, Looker, Advanced Excel
- **Databases & Warehousing:** PostgreSQL, MySQL, Google BigQuery, Snowflake, Amazon Redshift
- **Cloud & DevOps:** AWS (S3, EMR), Docker, Git
- **Gen AI & Machine Learning:** Scikit-learn, LLMs, RAG, Prompt Engineering

## WORK EXPERIENCE

**IT World Web** 12/2022 – 12/2024
*Executive Consultant – Business & Data* *Bengaluru, India*

- Built an **AI-assisted** QC validator for 25 leadership KPIs, cutting weekly validation from 2 hours to under 2 minutes through definition-based checks and AI-generated exception summaries.
- Led **UAT** for migration of 2M+ customer records to a cloud warehouse, identifying 400+ defects before launch, with zero critical issues in the first 90 days after cutover.
- Engineered a **funnel analytics pipeline** to identify conversion bottlenecks and analyze 500+ manual-review rules, driving automation that reduced review volume by 25%.

**Careernet Technologies** 04/2021 – 11/2022
*Associate Consultant – Business & Data* *Bengaluru, India*

- Investigated **230 reporting** discrepancies in six months, traced 52% to inconsistent business definitions, and drove remediation that reduced repeat exceptions by 34%.
- Engineered Amazon **QuickSight dashboards** to automate KPI tracking and operational reporting, reducing weekly Business Review preparation time by 12 hours.
- Defined **source-to-target** mappings and data validation rules, reducing reporting defects by 35% and shortening analytics turnaround by one week per cycle.

## PROJECTS

**AWS Event Analytics Lakehouse |** 🔗 03/2026 – Present
*Python, SQL, AWS, Airflow, Apache Iceberg*

- Architected an **Airflow**-orchestrated S3 to Iceberg pipeline using Athena SQL and incremental merges into date-partitioned Parquet tables, processing 50M events in 25 minutes.
- Reduced one-day replay scan volume from **11.5 GB** to **140 MB** and runtime 30% by implementing event-date partitioning and partition-aware merges.
- Automated infrastructure and CI using **Terraform** and GitHub Actions, with 49 regression tests validating quality gates, atomic snapshot publication, and lock recovery.

**Lakeflow Ingestion Framework** | 09/2025 – 04/2026

*Python, PySpark, Delta Lake, GitHub Actions*

- Built a metadata-driven **PostgreSQL to Databricks** framework for full loads and WAL-based CDC for 2M+ records across 4 tables, preserving INSERT, UPDATE, and DELETE history
- Implemented three-attempt exponential backoff and classified transient versus non-recoverable failures, reducing unnecessary retries and improving recovery from transient errors.
- Automated multi-environment **CI/CD** with GitHub Actions and Databricks Asset Bundles, enforcing linting, type checking, and a 70% test-coverage gate.

**GA4 Purchase Prediction Analytics** | 06/2025 – 10/2025

*Python, SQL, NumPy, Scikit-learn, Matplotlib, Docker*

- Built a GA4 purchase-prediction pipeline with BigQuery SQL and **Scikit-learn**, analyzing 4.3M events across 252K users with temporal validation.
- Accelerated feature aggregation 42× (11.84s → 0.28s) on 1M events using NumPy vectorization and categorical encoding, reducing DataFrame memory 80%.
- Strengthened batch inference with **SHA-256** model verification, atomic exports, and drift monitoring; automated regression tests and CLI checks.

**E-commerce Funnel Retention Analytics** | 01/2025 – 04/2025

*DuckDB, Python, SQL, Apache Parquet, pytest, GitHub Actions.*

- Built a **Python–DuckDB** analytics pipeline processing 1M events across 390K sessions, generating funnel and retention models in 16.2 seconds.
- Reduced median funnel-summary query time from 5.69 seconds to 1.81 milliseconds on 102K events by materializing SQL models and validating result equivalence.
- Engineered reliable ingestion with SHA-256 deduplication, quality gates, and atomic publication; automated 29 regression tests through GitHub Actions.

## CERTIFICATIONS & TRAININGS

- Analytical SQL for Developers – Oracle 07/2026
- Data Engineering on AWS Foundations – AWS 04/2026